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Studierichting: Informatica
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$$\lim_{x\to\infty}\left(\frac{x+\sqrt{2x^2+3x+2}}{2\sqrt{x^2-4x}-\sqrt[6]{-x^3+7x}}\right)$$

$$\lim_{x\to+\infty}\left(\sqrt[6]{-x^3+7x}\right)$$

$$= \lim_{x\to+\infty}\left(\sqrt[6]{x^3\left(-1+\frac{7}{x^2}\right)}\right)$$

$$= \sqrt[6]{(+\infty)\cdot\left(-1+\frac{7}{+\infty}\right)}$$

$$= \sqrt[6]{(+\infty)\cdot(-1+0)}$$

$$= \sqrt[6]{-\infty}$$

Limiet is niet gedefinieerd, want $-\infty$ heeft geen zesde machtswortel.

$$\lim_{x\to-\infty}\left(\frac{x+\sqrt{2x^2}}{2\sqrt{x^2}-\sqrt[6]{(-x)^3}}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2x^2}}{2\sqrt{x^2}-(-x)^{\frac{3}{6}}}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2x^2}}{2\sqrt{x^2}-(-x)^{\frac{1}{2}}}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2x^2}}{2\sqrt{x^2}}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2}\cdot\sqrt{x^2}}{2\sqrt{x^2}}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2}\cdot|x|}{2\cdot|x|}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x+\sqrt{2}\cdot(-x)}{2\cdot(-x)}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x-x\sqrt{2}}{-2x}\right)$$

$$= \lim_{x\to-\infty}\left(\frac{x(1-\sqrt{2})}{-2x}\right)$$

$$= \frac{1-\sqrt{2}}{-2}$$

$$= \frac{(-1) \cdot (1 - \sqrt{2})}{(-1) \cdot (-2)}$$
$$= \frac{\sqrt{2} - 1}{2}$$

References